

Kunal Maheshwari

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SUMMARY

OPT-EAD holder with full-time work authorization, bringing over 4 years of professional experience as a Software Developer and Java Engineer. Possesses a Master of Science in Computer Science with expertise in building scalable microservices, distributed data pipelines, and backend systems. Demonstrates a strong command of Java, Python, Spark, SQL, AWS, and Azure, coupled with experience in CI/CD automation, containerization with Docker, and graduate-level work on large-scale data platforms and privacy-preserving AI applications. Proficient in backend development, distributed systems, and orchestration tools.

EDUCATION

Central Michigan University | Mount Pleasant, MI Aug 2024 - Dec 2025
Master of Science | Computer Science

Lovely Professional University | Punjab, India Aug 2016 - May 2020
Bachelor of Technology | Computer Science and Engineering

EXPERIENCE

Software Developer | NTT DATA | Bengaluru, India (Remote) August 2022 - August 2024

- Designed and optimized high-performance REST and SOAP microservices using Java 8/11 and Spring Boot, handling over 50,000 daily API requests with 99.99 percent SLA uptime.
- Engineered and automated CI/CD pipelines with Jenkins and Azure DevOps, facilitating continuous delivery across Azure Cloud and Oracle WebLogic environments.
- Modernized legacy systems by migrating to Log4j 2.x, improving monitoring and error handling, which reduced issue resolution time by 20 percent.
- Spearheaded a major EDX platform upgrade, streamlining application architecture and dependency management to achieve 50 percent faster delivery cycles.
- Collaborated within Agile and Scrum teams to deliver FHIR-compliant healthcare solutions using Git, SVN, and comprehensive JUnit testing.

Senior Analyst | Capgemini | Mumbai, India (Remote) October 2021 - August 2022

- Engineered robust Java and EJB-based enterprise applications and RESTful web services for prominent automotive clients.
- Enhanced system performance by 90 percent through meticulous SQL query optimization, database tuning, and modernization of legacy Java code.
- Implemented CI/CD pipelines using Jenkins and GitLab, integrating automated testing and quality gates for reliable deployments.
- Mentored junior developers by conducting thorough code reviews, facilitating debugging sessions, and promoting clean coding practices.

Analyst | Capgemini | Mumbai, India (Remote) July 2020 - October 2021

- Maintained high-availability enterprise systems with over 99 percent uptime through proactive monitoring and performance optimization.
- Enhanced SQL data pipelines and database operations, improving query performance by 10 percent and strengthening data integrity.
- Implemented comprehensive unit testing using JUnit4, achieving 99 percent code coverage for critical modules.
- Supported release planning and sprint execution using JIRA within Agile development teams, ensuring timely project delivery.

PROJECTS

NYU Taxi Data Analysis Platform January 2024 - May 2025

- Built a distributed Spark-based data pipeline to process and analyze over 8 GB of NYC taxi trip data in near real-time.
- Developed an interactive Flask web application for data exploration with dynamic filtering and visualization capabilities.
- Containerized ingestion, processing, and analytics components using Docker for reproducible and scalable deployments.
- Optimized Spark memory usage and partitioning strategies, reducing overall job runtime by 40 percent.
- Automated cloud infrastructure provisioning using Terraform across AWS services including S3, RDS, ECS, Lambda, and QuickSight.

Privacy-Preserving AI Email Client (Capstone Project) January 2025 - December 2025

- Engineered a fine-tuning pipeline for T5-based transformer models to perform advanced email summarization and classification tasks.
- Applied parameter-efficient fine-tuning using LoRA to support low-resource training under stringent privacy constraints.
- Implemented model evaluation using ROUGE metrics and integrated quantization for secure and efficient local deployment.

SKILLS

- Programming Languages: Proficient in Java, Python, and SQL for robust backend development and data manipulation. Familiar with C++ and Javascript/Typescript.
- Frameworks and Libraries: Expertise in Spring Boot, Spring Framework, Apache Spark, PySpark, Hugging Face, T5, and LoRA for building scalable applications and AI models. Skilled in Grpc and GraphQL for efficient communication.
- Web Services & APIs: Extensive experience with REST API and SOAP API development. Experience with GraphQL for flexible data querying.
- Cloud Platforms & Tools: Hands-on experience with AWS (EC2, RDS, S3, Lambda), Azure (Functions, DevOps, ServiceBus), and Google Cloud Platform. Utilized containerization technologies like Docker and Kubernetes for orchestration.

- **Databases:** Proficient with PostgreSQL, Oracle Database, IBM DB2, and AWS RDS. Experienced with Splunk for log analysis and monitoring.
- **Development Tools & Concepts:** Skilled in Jenkins, Git, SVN, Maven, JIRA, Confluence, Tableau, DBeaver, and Terraform for CI/CD, version control, project management, and infrastructure automation. Deep understanding of ETL Pipelines, Data Warehousing, Data Analytics, Agile, Scrum, Test Driven Development, Object Oriented Programming, CI/CD, Microservices, Distributed Systems, Large-scale data platforms, Infrastructure automation, and Privacy-preserving LLM-based applications.
- **Monitoring & Visualization:** Familiar with Grafana and Prometheus for system monitoring, and Tableau for data visualization.